

GLEICHEN AGCM, AB, Canada

WMO: 714980

Lat: 50.9345N

Lon: 112.9408W

Elev: 907

StdP: 90.89

Time Zone: -7.00 (NAM)

Period: 08-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-29.2	-26.3	-32.4	0.2	-29.2	-29.3	0.3	-26.1	12.5	-2.3	11.2	-2.8	2.9	210	0.669

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.1	29.5	16.4	27.7	15.7	26.0	15.3	18.7	25.5	17.7	24.7	16.7	23.6	3.7	150

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
16.3	12.9	21.9	15.0	11.9	21.0	13.9	11.1	19.6	56.9	26.0	53.4	24.7	50.3	23.6	24.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 11.2	9.5	8.4	DB	-32.7	33.0	2.4	2.0	-34.4	34.4	-35.7	35.6	-37.1	36.7	-38.8	38.2	(4)
(5)			WB	-32.8	21.4	2.3	1.3	-34.5	22.4	-35.8	23.1	-37.1	23.8	-38.8	24.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	4.0	-8.0	-9.6	-3.1	4.1	10.7	14.6	17.2	16.7	12.1	4.9	-2.9	-9.6	(6)	
	DBStd	11.56	8.13	8.39	8.15	4.98	4.40	3.06	2.64	3.16	4.77	5.26	7.40	8.06	(7)	
	HDD10.0	2945	558	548	407	183	46	3	0	1	32	171	389	607	(8)	
	HDD18.3	5279	816	781	665	426	239	118	54	69	191	417	638	865	(9)	
	CDD10.0	752	0	0	0	7	66	140	222	207	96	13	2	0	(10)	
	CDD18.3	46	0	0	0	0	1	5	17	17	5	0	0	0	(11)	
	CDH23.3	1088	0	0	0	8	78	117	343	376	162	6	0	0	(12)	
	CDH26.7	267	0	0	0	1	11	21	84	109	42	0	0	0	(13)	
(14) Wind	WSAvg	3.7	4.2	3.8	4.1	4.2	4.3	3.9	2.9	2.7	3.4	3.7	3.8	3.8	(14)	
(15) Precipitation	PrecAvg	362	14	10	15	26	49	64	52	40	44	13	15	14	(15)	
	PrecMax	510	49	26	40	75	110	163	153	93	158	37	74	39	(16)	
	PrecMin	243	0	0	3	1	7	7	14	6	3	0	1	0	(17)	
	PrecStd	76	10	6	10	22	26	38	32	24	35	10	13	9	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.2	9.4	16.3	23.3	28.2	29.1	31.5	32.0	30.5	23.5	17.9	9.5	(19)	
		MCWB	3.5	4.3	7.7	10.8	12.9	16.1	18.7	16.1	14.8	11.7	8.7	4.1	(20)	
	2%	DB	5.9	5.8	13.3	19.6	25.6	26.4	29.0	29.5	27.8	19.2	12.6	5.7	(21)	
		MCWB	2.5	2.3	5.9	8.8	12.3	15.0	17.8	16.6	13.8	10.0	5.9	2.3	(22)	
	5%	DB	4.4	3.5	10.2	16.2	23.1	24.3	27.0	27.5	24.9	16.5	9.2	3.6	(23)	
		MCWB	1.8	0.8	4.4	6.7	11.1	14.2	17.3	16.1	13.2	8.5	4.4	0.8	(24)	
	10%	DB	2.9	1.4	6.9	13.3	20.5	22.5	25.1	25.3	21.9	13.6	6.5	1.4	(25)	
		MCWB	0.6	-0.4	2.8	5.4	10.1	13.4	16.7	15.5	12.1	7.1	2.4	-0.8	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	4.2	4.9	8.3	11.0	13.7	18.3	21.2	19.2	16.0	12.6	8.7	4.6	(27)	
		MCDB	7.5	8.3	14.9	22.3	25.0	25.1	27.8	26.2	26.8	22.0	16.8	8.0	(28)	
	2%	WB	2.7	2.4	6.4	9.0	12.8	16.6	19.3	18.1	14.6	10.6	6.5	2.5	(29)	
		MCDB	5.4	5.4	12.4	18.2	23.3	23.7	26.0	26.1	24.9	17.7	11.8	5.8	(30)	
	5%	WB	1.9	1.0	4.6	7.3	11.7	15.3	18.2	17.2	13.5	9.0	4.7	0.9	(31)	
		MCDB	4.3	3.2	9.6	14.9	21.4	21.8	24.6	25.1	23.2	15.7	8.9	3.6	(32)	
	10%	WB	0.6	-0.4	3.0	5.9	10.7	14.3	17.3	16.2	12.5	7.6	2.8	-0.7	(33)	
		MCDB	2.9	1.3	6.7	12.3	18.7	20.7	23.7	23.3	21.3	13.1	5.8	1.3	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	9.4	9.8	10.8	12.7	13.7	12.8	14.1	14.6	14.4	12.8	10.4	9.2	(35)	
		MCDBR	9.4	11.6	15.1	18.4	18.4	16.3	17.9	19.0	19.9	17.2	14.3	10.9	(36)	
	5% WB	MCWBR	7.0	8.6	8.8	9.1	7.5	7.1	8.7	8.0	8.1	8.9	8.6	8.2	(37)	
		MCWBR	9.4	11.1	14.1	17.3	16.1	14.3	15.8	16.3	18.1	16.5	13.9	11.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.260	0.266	0.285	0.338	0.359	0.351	0.351	0.368	0.323	0.282	0.254	0.238	(40)	
		taud	2.389	2.451	2.468	2.323	2.318	2.387	2.408	2.344	2.441	2.533	2.490	2.414	(41)	
	Edn at Noon		788	887	929	902	895	904	898	861	870	857	800	768	(42)	
		Edh at Noon	60	74	88	114	121	114	110	111	90	67	53	49	(43)	
(44) All-Sky Solar Radiation	RadAvg		1.17	2.17	3.51	4.79	5.75	6.24	6.74	5.41	3.82	2.34	1.26	0.89	(44)	
	RadStd		0.09	0.19	0.28	0.35	0.40	0.39	0.33	0.36	0.37	0.27	0.11	0.08	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (82 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page